over all HSC filters). We note that this does approximate the sensitivity of each HSC filter as a top-hat, but this should be adequate as the transmittance of each HSC filter is mostly uniform over their respective range (Kawanomoto et al. 2018). Any error introduced will be significantly smaller than the dispersion in the stacked SEDs (see Fig. 5)

We stress here that the HSC flux-densities used are the *decomposed* AGN fluxes from Li et al. (2024), so do not contain host-galaxy contamination. These flux-densities are then simply converted to luminosity densities through the luminosity distance, d_L, of each source, calculated from their redshift using our assumed cosmology (Planck Collaboration et al. 2020).

2.3.2 eROSITA

The eROSITA data need to be unfolded from the instrument response before we can re-bin them onto a common grid. For this we use XSPEC v.12.13.0c (Arnaud et al. 1985)

For each eROSITA spectrum we apply a simple absorbed power-law model using TBABS*POW, where the galactic column in the absorption model TBABS is fixed at that from HI4PI Collaboration et al. (2016), and POW gives the power-law component, where we use the values derived by Liu et al. (2022). We stress that unlike Liu et al. (2022), we *do not* include an intrinsic absorption component to model any potential absorption from within the AGN itself (i.e they used TBABS*ZTBABS*POW, while we omit the ZTBABS component). We neglect the intrinsic absorption component because our sample, by design, should only include unobscured sources, and hence we assume the observed spectral shape after correcting for Galactic absorption is representative of the true SED. This assumption is tested in detail in Appendix B

We use the above model to unfold the raw data from the instrument response, noting that it makes no assumptions on any spectral complexity beyond a power-law, providing raw flux densities. We then remove the absorption component from the model, and write out the unabsorbed model. The energy dependent absorption correction factor for each eROSITA energy bin is simply the ratio of the unabsorbed to absorbed model, which is then applied to the raw flux-densities to give the galactic absorption corrected flux-densities.

The absorption corrected flux densities are then corrected for redshift and re-binned onto a common energy grid. eROSITA extends from 0.2 keV to 8 keV. Since the sensitivity of eROSITA drops off rapidly at higher energies (Predehl et al. 2021), we choose to limit the upper observed frame energy to 5.3 keV. For our redshift range this gives rest-frame energy limits from 0.36 keV to 9.6 keV. We use these to define a common rest-frame energy grid, with bin edges at [0.4, 0.7, 1, 2, 4, 8] keV. These bins are intentionally wider than the intrinsic eROSITA bins in order to simplify the rebinning process and maximise signal-to-noise by providing higher photon counts per bin. First we find all original energy bins (shifted to the AGN rest-frame) that lie within the new energy bin. We then calculate the log-mean flux density of all original bins that lie within the new bin, using the absorption corrected flux-densities. This then provides the flux-density within the new energy bin.

As with the HSC flux-densities, these are converted to luminosity densities through the luminosity distance, again using the source redshift and Planck 2018 Cosmology results (Planck Collaboration et al. 2020).

2.4 Redshift Distribution

The left panels of Figure 2 show the redshift distribution of our sample for 3500 Å monochromatic luminosity (*top*) and 2 keV monochromatic luminosity (*bottom*) respectively, using the redshift measured in Salvato et al. (2022). The right panel of each figure shows the distribution for the subset of sources, with black hole masses within $\log_{10} M/M_\odot \in [8.0, 8.5]$, which we use for further analysis (see section 2.5).

The source numbers (top histogram) are relatively evenly distributed with redshift in both 3500 Å luminosity and 2 keV luminosity in both the full sample and the $\log_{10} M/M_\odot \in [8.0, 8.5]$ mass bin apart from a drop in numbers below $z \leq 0.25$ due to the small volume. Similarly, the mean source luminosity increases with redshift as the larger volume includes the rarer, more luminous objects.

2.5 Mass and Luminosity Distribution

Each AGN within our sample now has a rebinned, deabsorbed rest frame spectrum with a black hole mass estimate. We start by subdividing our sample into black hole mass bins, from $\log_{10} M/M_\odot = 4$ to $\log_{10} M/M_\odot = 9.5$ with bin width 0.5 dex (note that we only show bins with $\log M/M_\odot \geq 6$ in Fig. 3 due to limited sources below this). This allows us to control for changes in the SED due to differences in black hole mass leading to intrinsic differences in the temperature of the accretion flow (i.e a larger black hole gives a cooler accretion disc; Shakura & Sunyaev 1973). We have chosen relatively broad bins of 0.5 dex to allow for the intrinsic uncertainty in the black hole mass estimates. Fig. 1 show that the majority of our mass estimates (from the host galaxy stellar mass) lie within 0.5 dex of the single epoch mass estimates from SDSS DR 16 (Wu & Shen 2022) (which itself has an intrinsic uncertainty). Hence we do expect there to be some leakage between mass bins, however by choosing broad bins the overall effect of this on the stacked SEDs should be minimal.

Within each mass-bin we further subdivide the sample into bins of monochromatic optical luminosity, as this should originate from the disc and as such trace the mass-accretion rate, which significantly impacts the shape of the SED (Kubota & Done 2018). We choose the monochromatic luminosity at 3500 Å, as the bluer the wavelength the closer to the SED peak it will originate from, providing a better tracer of mass-accretion rate. This leads to a 2D-distribution, visualised in Fig. 3 Here each mass-luminosity grid is shown as a green rectangle, where the shade indicates the number of sources within each bin (with a darker green indicating more sources, as shown by the colour-bar). We note that though a regular grid is used for to visualise to overall sample, in the SED analysis later we optimise the luminosity bin edges for each mass-bin separately to give contiguous bins with an even distribution throughout each luminosity bin (in order to avoid side-weighted distributions in the edge bins). The black-grey contours show the distribution of our sources that are also included in the SDSS DR16 AGN catalogue (Wu & Shen 2022). These are objects selected for spectroscopic followup based on their photometric colours showing a blue continuum (Richards et al. 2002), and these spectra show the characteristic broad emission lines. This immediately shows that our X-ray selected sample extends down to considerably lower optical AGN luminosity than that of the SDSS AGN catalogue. This is more clear in Fig. 4, which shows the 3500 Å luminosity distribution for a single mass bin, $\log_{10} M/M_\odot \in [8.0, 8.5]$, for both the full sample (green, solid) and the sources matched to the SDSS AGN catalogue (blue, hatched). Here it is clear that not only does our sample extend to considerably lower optical luminosity, it also peaks at lower luminosity, indicat-

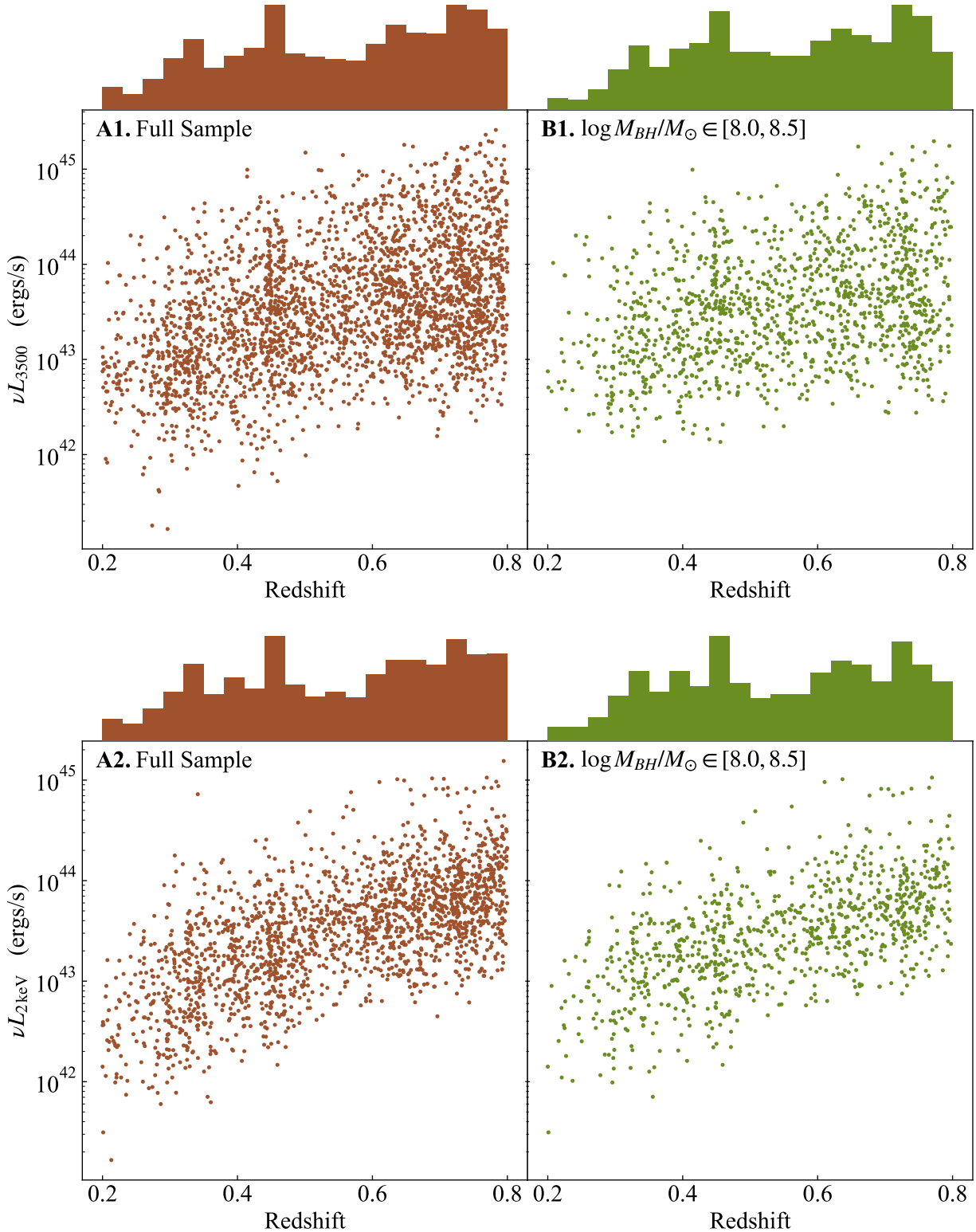

Figure 2. Redshift and monochromatic luminosity distributions for 3500 Å (*top*) and 2 keV (*bottom*). In both cases the left panel shows the full sample and the right the sub-sample within our fiducial $\log_{10} M/M_\odot \in [8.0, 8.5]$ mass-bin. The top histograms show the redshift distribution of the samples, normalised such that their integral is unity. It is clear that the distributions do not change significantly between the full sample and our fiducial mass-bin. In all cases there is a slight deficit in the number of sources below $z \sim 0.25$, however the remaining redshift range (up to $z = 0.8$) has a relatively even distribution.

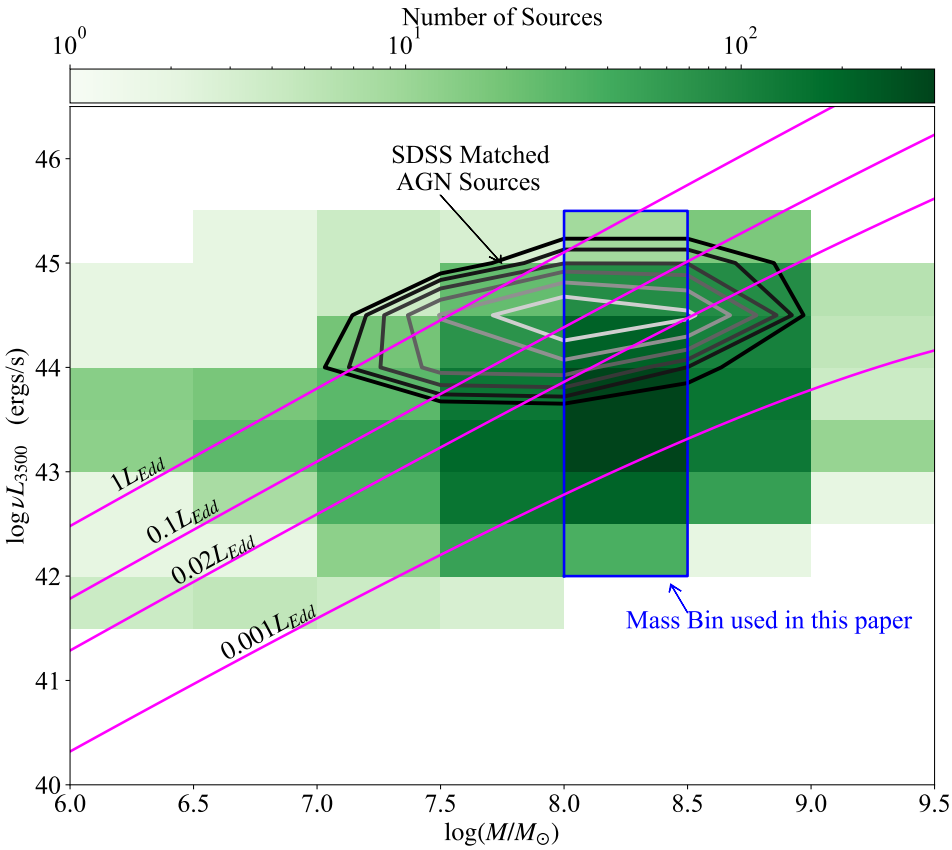

Figure 3. Distribution of our sample across bins of black hole mass and monochromatic 3500 Å luminosity. The green shading gives the number of sources per bin (as defined by the colour-bar on top). The black contour lines show the distribution of sources that have matches with the SDSS DR16 AGN catalogue (Wu & Shen 2022). Magenta lines show the predicted 3500 Å luminosity at different Eddington ratios, calculated for a standard accretion disc extending to the ISCO, while the blue box indicates the bins used for the main analysis presented in this paper. It is clear that our eROSITA-HSC combined sample pushes to significantly lower optical luminosity than the broad-line SDSS sources, allowing us to probe AGN accretion below the 2 % of Eddington transition.

ing that bright broad-line AGN are not necessarily the dominant unobscured AGN population.

A more interesting metric than the monochromatic luminosity is the Eddington ratio. This is shown by the magenta lines in Fig. 3. We stress here that these are calculating from converting the 3500 Å luminosity to bolometric luminosity via a standard Shakura & Sunyaev (1973) accretion disc model, and as such the precise values should be treated with caution. For $L/L_{\rm Edd} \lesssim 0.02$, where we see the changing-look transition, we expect a vastly different accretion structure (Done et al. 2007; Noda & Done 2018), and so converting via a disc model is not accurate. However, it is sufficient to give an estimate of what distribution of sources lie above or below the 2 % Eddington transition. It is clear that the majority of the SDSS AGN sources lie above this transition, as seen previously in Mitchell et al. (2023), while the majority of our sample lie below (for black hole masses $\log_{10} M/M_\odot \gtrsim 7$ - below this we do not trust the mass estimates). Again this becomes more evident in Fig. 4, where we see that our sample also peaks below the 2 % transition. This already gives strong evidence for a change in SED at low Eddington ratios, where the optical luminosity drops much faster than the X-ray luminosity, as predicted if there is a transition in the physical nature of the accretion flow from the optical being dominated by a disc to an ADAF.

We investigate this in more detail by stacking the spectra within each bin, averaging in logarithmic luminosity on the common wavelength/energy binning. This gives an average SED in each mass-luminosity bin. For the remainder of the paper we focus on the single mass bin $\log_{10} M/M_\odot \in [8.0, 8.5]$, as this is the bin where the black hole mass estimates from the host galaxy are most reliable, and also where we have most objects. We note here that the following analysis is also performed for the bins $\log_{10} M/M_\odot \in [7.5, 8.0]$ and $[8.5, 9.0]$ as a consistency check. These are presented in Appendix C). Using a single mass bin gives a sequence of SEDs for changing monochromatic 3500 Å luminosity at a fixed mass, shown in Fig. 5 (orange (HSC) and blue (eROSITA) points), where the luminosity bins are chosen to precisely fit around the underlying luminosity distribution, giving bins of width 0.4 dex, each with an even luminosity distribution within, as shown by the vertical black dotted lines in Fig. 4. It is immediately clear that the 3500 Å monochromatic luminosity changes by over 3 orders of magnitude while the X-ray power changes by less than a factor of a few. It is also immediately clear that the shape of the optical spectrum changes dramatically, from a blue continuum like a standard disc in the higher luminosity bins, to a fairly flat continuum in the middle bins, to a strongly red continuum very unlike a standard disc in the lowest luminosity bins. Again, this suggests a dramatic change in the accretion structure, which we explore in greater detail in the next section (3), where we detail our SED model and apply it to the stacked SEDs to give a physical interpretation of the change in SED shape.